Tracking Satellites 'Over the Top'

A Practical Rotator Control Technique for Hemispherical Pass Anomalies

Introduction

Amateur satellite operators routinely track low Earth orbit (LEO) satellites using azimuth-elevation (AZ/EL) rotator systems. In most passes, the rotator simply follows the satellite across the sky without incident. However, a particular class of pass presents a serious mechanical challenge: those in which the satellite crosses the rotator's 'wrap-around' boundary — the azimuth heading at which the rotator must reverse direction or rotate nearly a full 360 degrees to continue tracking.

This article describes the problem in detail, explains why it occurs differently depending on your geographic location, and presents a practical software technique — the 'over-the-top' method — that eliminates the issue entirely, provided your elevation rotator supports movement beyond 90 degrees.

The Wrap-Around Problem

Understanding Rotator Types

Azimuth rotators are manufactured in two common variants, and which type you are likely to own depends largely on which hemisphere you live in.

Property	North-Centered (Typically SH)	South-Centered (Typically NH)
Range	-180° → 0° → +180°	0° → 360° (or 0° → 180° → 360°)
Full CCW position	-180° (South)	0° (North)
Wrap-around boundary	±180° (South)	0°/360° (North)
Problematic passes	Passes crossing South (e.g., AOS=225°, LOS=135°)	Passes crossing North (e.g., AOS=330°, LOS=90°)

What Happens During a Problematic Pass

Consider a Southern Hemisphere operator using a south-centered rotator. A satellite rises in the southwest at Az=225°, climbs to a peak near Az=180° (due south), and sets in the southeast at Az=135°. The rotator tracks perfectly until the satellite approaches 180°. To continue tracking, the controller would need to move from, say, Az=181° to Az=179° — but the rotator's mechanical range ends at ±180°. To reach 179°, the drive motor must reverse direction and travel nearly 360° clockwise (past north, east, south, and back). On a typical rotator with a slew speed of around 4 degrees per second, this journey takes well over a minute.

During that minute, the antenna is pointed nowhere near the satellite. For a voice or data contact, this is catastrophic. The link drops, the pass is lost, and the opportunity is wasted.

The equivalent problem for Northern Hemisphere operators occurs when a satellite passes to their north — rising at Az=330° and setting at Az=90°, crossing through Az=0°/360°. The mechanics are identical, and the solution is the same.

The Over-the-Top Solution

Elevation Beyond 90 Degrees

Most amateur-grade elevation rotators designed for satellite work support elevation movement not just from 0° (horizon) to 90° (zenith), but all the way to 180°. An elevation of, for example, 135° means the antenna is tilted past vertical and is now pointing toward a direction 45° above the horizon on the opposite side of the rotator.

This mechanical capability is the key to the over-the-top technique.

The Geometric Principle

The insight is straightforward: any point in the sky can be reached from two different azimuth headings. If a satellite is at Az=180°, El=35°, you can point your antenna at it directly — or you can point your antenna at Az=0°, El=145°. In the latter case, the antenna is rotated to face north, but elevated past vertical so that it points back over the top toward the south at an angle of 35° above the southern horizon.

The transformation is:

> **Real Azimuth = (Satellite Azimuth + 180°) mod 360°**
> **Real Elevation = 180° − Satellite Elevation**

Applied continuously during the pass, this means the azimuth rotator is smoothly tracking through the opposite side of the sky — through its comfortable travel range near Az=0° (or Az=180° for the Northern Hemisphere case) — while the elevation rotator holds the antenna pointed at the correct elevation from the other side.

When to Apply the Technique

The over-the-top transformation does not need to be applied for the entire pass — only when the satellite is in the half of the sky where the wrap-around problem would otherwise occur. Tracking software can make this decision automatically by checking the current satellite azimuth against the rotator type and switching seamlessly between direct and over-the-top modes.

Pre-Pass Planning: Deciding When to Use Over-the-Top

Why the Decision Cannot Be Made Mid-Pass

A critical implementation detail — one that is easy to overlook — is that the decision to use over-the-top mode cannot safely be made on the fly during a pass. It must be made in full, before AOS, based on a pre-computed prediction of the complete azimuth track.

The reason is mechanical. Consider what would happen if the software detected the wrap-around problem only after the satellite had already risen and tracking had begun in direct mode. At the moment the software switched to over-the-top, the antenna azimuth would need to jump by

approximately 180 degrees — for example, from Az=170° to Az=350°. This is precisely the kind of large, sudden slew that the over-the-top technique is designed to avoid. Switching modes mid-pass does not solve the problem; it creates an equally disruptive slew at a different point in the pass.

The correct approach is to compute the complete predicted pass — typically at one- to five-second intervals from AOS to LOS — before the satellite rises, analyse the full azimuth trajectory for a wrap-around crossing, make the mode decision once, and then apply it consistently for the entire pass. The antenna is then pre-positioned to the correct starting position (direct or over-the-top) before the satellite appears above the horizon.

The Pass-Planning Algorithm

Software developers implementing this feature should structure their tracking engine around the following workflow:

Step 1 — Predict the full pass trajectory: Using the orbital elements (TLE) and a propagator (SGP4 or equivalent), compute the predicted azimuth and elevation of the satellite at regular intervals — typically every 1 to 5 seconds — from AOS (elevation = 0° rising) to LOS (elevation = 0° setting). Store this as a list of (Az, El, time) tuples. This is the complete azimuth track for the pass.

Step 2 — Detect a wrap-around crossing: Scan the predicted azimuth track and determine whether it crosses the rotator's wrap-around boundary. For a north-centered rotator (Southern Hemisphere), the boundary is at Az=180°: a crossing is present if the track contains azimuths on both sides of 180° (i.e., some values in the 90°–180° range and others in the 180°–270° range). For a south-centered rotator (Northern Hemisphere), the boundary is at Az=0°/360°: a crossing is present if the track contains azimuths both above 270° and below 90°, which can only occur by passing through North.

Step 3 — Decide the mode for the whole pass: If a wrap-around crossing was detected, set use_over_top = True for this pass. If no crossing was detected, set use_over_top = False. This flag applies to every tracking point in the pass without exception. It is not re-evaluated during the pass.

Step 4 — Pre-position the antenna before AOS: Using the decided mode and the first point in the predicted trajectory (the AOS position), compute the starting antenna angles: if direct mode, point to the AOS azimuth and elevation directly; if over-the-top mode, point to (AOS_Az + 180°) mod 360° and elevation (180° − AOS_El). Since AOS elevation is zero or near-zero, the over-the-top pre-position will have an elevation close to 180°. Drive the antenna to this position before the satellite rises, so no slewing is needed once tracking begins.

Step 5 — Track continuously using the pre-decided mode: As the satellite rises and tracking begins, apply the decided transformation (or no transformation) to every AZ/EL update from the propagator and send the resulting real angles to the rotator controller. The mode does not change until LOS.

Detecting the Wrap-Around Crossing: A Closer Look

The wrap-around detection in Step 2 deserves particular attention, because the azimuth values reported by satellite propagators do not inherently signal a crossing — they simply continue numerically. The developer must inspect the spread of azimuth values across the predicted pass.

For a north-centered rotator, the question is: does the azimuth track span across Az=180°? This is true when the track contains values approaching from the east (90°–180°) and values departing to the west (180°–270°), or vice versa. Simply checking whether any point equals 180° is not robust — the prediction interval may skip over exactly 180°. Checking for values on both sides of 180° is the correct approach.

For a south-centered rotator, the question is: does the azimuth track span across Az=0°/360°? Because azimuth is typically reported in the range 0–360, a track that crosses north will contain values near 350°–360° and also values near 0°–10°. The crossing is detected when the track contains values both above 270° and below 90°. Note that a pass entirely in the southern sky might also contain values in both those ranges (e.g., rising at 200°, setting at 160°, with peaks dipping toward Az=90° or Az=270°), so it is important to check for the simultaneous presence of high and low values, not either alone.

One additional edge case worth handling: a pass that rises exactly on the wrap-around boundary (AOS Az ≈ 180° for a north-centered rotator, or AOS Az ≈ 0° for a south-centered rotator). In this situation there may be no values on the 'approaching' side, yet the pass will immediately encounter the problem zone. A conservative implementation adds a small guard margin — for example, treating any pass with AOS within ±10° of the boundary as requiring over-the-top mode regardless of whether values on both sides are present.

Passes That Do Not Require Over-the-Top

Not every pass requires the over-the-top technique, and it should not be applied unnecessarily. Using over-the-top when it is not needed forces the elevation rotator to hold a high angle (near 180°) for an entire pass, which places unnecessary mechanical load on the drive system and, for linearly polarised antennas, inverts the polarisation plane for no benefit.

For a Southern Hemisphere station with a north-centered rotator, passes that rise and set entirely in the northern half of the sky (e.g., AOS=45°, LOS=315°, transiting through North) pose no wrap-around problem at all — the rotator simply tracks from one side of north to the other within its comfortable range. These passes should be tracked in direct mode. The same logic applies to a Northern Hemisphere station for passes that remain entirely in the southern sky.

The pass-planning algorithm described above handles this automatically: if no wrap-around crossing is detected in the predicted trajectory, direct mode is used and the antenna is pre-positioned to the direct AOS coordinates.

Python Implementation

The following Python code implements the complete workflow described above. It is structured in two parts: first the pass-planning functions (which must run before AOS), then four simulated pass scenarios demonstrating both cases — passes that require over-the-top and passes that do not — for both rotator types.

```
import math

# ======================================================================
# PART 1: Pass Planning — must be run BEFORE the pass begins
```

```
# ============================================================================

def crosses_wrap_boundary(pass_points, rotator_type):
    """
    Analyse a complete predicted pass and determine whether the satellite's
    azimuth track will cross the rotator's wrap-around boundary.

    pass_points : list of (az_deg, el_deg) tuples covering the full pass,
                  typically at 1–5 second intervals from AOS to LOS.
    rotator_type: 'north_centered' (SH stations, range -180..0..+180)
                  'south_centered' (NH stations, range 0..180..360)

    Returns True if the wrap boundary is crossed during this pass.
    """
    if rotator_type == 'north_centered':
        # Wrap boundary is at Az = 180 (South).
        # A crossing is detected when the azimuth track contains points
        # on BOTH sides of 180 degrees — i.e. some points in the eastern
        # half (90..180) and some in the western half (180..270).
        has_east_approach  = any(90  < az < 180 for az, el in pass_points)
        has_west_approach  = any(180 < az < 270 for az, el in pass_points)
        return has_east_approach and has_west_approach
    else:
        # Wrap boundary is at Az = 0 / 360 (North).
        # A crossing is detected when the track contains azimuths both
        # above 270 (approaching from the west) and below 90 (departing
        # to the east), which can only happen by crossing through North.
        has_west_approach  = any(az > 270 for az, el in pass_points)
        has_east_departure = any(az <  90 for az, el in pass_points)
        return has_west_approach and has_east_departure


def plan_pass(pass_points, rotator_type):
    """
    Decide — BEFORE the pass begins — whether to use over-the-top mode
    for the entire pass.

    This function MUST be called with the complete predicted trajectory
    before AOS.  The mode cannot safely be changed mid-pass: switching
    from direct to over-the-top (or vice versa) while the antenna is
    already tracking would require a large, sudden azimuth slew that
    would lose the satellite.

    Returns a dict:
      {
        'use_over_top': bool,   # True = use OTT for the whole pass
        'reason':      str,    # Human-readable explanation
        'preposition':  (az, el) or None   # Where to point before AOS
      }
    """
    needs_ott = crosses_wrap_boundary(pass_points, rotator_type)

    if needs_ott:
        # Pre-position the antenna to the OTT equivalent of the AOS point
        aos_az, aos_el = pass_points[0]
        pre_az = (aos_az + 180) % 360
        pre_el = 180 - aos_el        # Will be ~180 deg since el~0 at AOS
        return {
            'use_over_top': True,
            'reason': 'Pass crosses the wrap-around boundary — OTT required.',
            'preposition': (pre_az, pre_el)
        }
```

```
    else:
        aos_az, aos_el = pass_points[0]
        return {
            'use_over_top': False,
            'reason': 'Pass does not cross the wrap-around boundary — direct tracking.',
            'preposition': (aos_az, aos_el)
        }


def compute_antenna_angles(sat_az_deg, sat_el_deg, use_over_top):
    """
    Apply (or not) the over-the-top transformation for a single tracking
    point, given the mode that was decided before the pass started.

    sat_az_deg  : satellite azimuth (0–360)
    sat_el_deg  : satellite elevation (0–90)
    use_over_top: bool — set once per pass by plan_pass(), never toggled
                  during the pass.
    """
    if use_over_top:
        real_az = (sat_az_deg + 180) % 360
        real_el = 180 - sat_el_deg
        method  = "Over-the-Top"
    else:
        real_az = sat_az_deg
        real_el = sat_el_deg
        method  = "Direct"

    return real_az, real_el, method


# ============================================================================
# PART 2: Simulated passes — demonstrating the full workflow
# ============================================================================

def run_simulation(label, pass_points, rotator_type):
    print("=" * 70)
    print(f"  {label}")
    print("=" * 70)

    # --- Step 1: Plan the pass BEFORE AOS ---
    plan = plan_pass(pass_points, rotator_type)
    print(f"  Rotator type : {rotator_type}")
    print(f"  OTT required : {plan['use_over_top']}")
    print(f"  Reason       : {plan['reason']}")
    az0, el0 = plan['preposition']
    print(f"  Pre-position : Az={az0:.1f} deg  El={el0:.1f} deg  (before AOS)")
    print()
    print(f"{'Time':>6} {'Sat Az':>8} {'Sat El':>8} {'Real Az':>8} {'Real El':>8} {'Method'}")
    print("-" * 60)

    # --- Step 2: Track the pass using the pre-decided mode ---
    for i, (az, el) in enumerate(pass_points):
        t = i * 30
        real_az, real_el, method = compute_antenna_angles(
            az, el, plan['use_over_top']
        )
        print(f"{t:>5}s {az:>8.1f} {el:>8.1f} {real_az:>8.1f} {real_el:>8.1f} {method}")

    print()
```

```
# ------------------------------------------------------------------------
# Scenario A: Southern Hemisphere — pass transits South (OTT required)
# ------------------------------------------------------------------------
sh_pass_south = [
    (225, 0), (210, 8), (195, 18), (183, 28),
    (180, 35), (177, 28), (165, 18), (150, 9), (135, 0),
]
run_simulation(
    "Southern Hemisphere — satellite transits South (OTT required)",
    sh_pass_south, 'north_centered'
)

# ------------------------------------------------------------------------
# Scenario B: Southern Hemisphere — pass stays in the northern sky (no OTT)
# ------------------------------------------------------------------------
sh_pass_north = [
    (45, 0), (30, 12), (10, 25), (355, 40),
    (340, 25), (325, 12), (315, 0),
]
run_simulation(
    "Southern Hemisphere — satellite transits North (no OTT needed)",
    sh_pass_north, 'north_centered'
)

# ------------------------------------------------------------------------
# Scenario C: Northern Hemisphere — pass transits North (OTT required)
# ------------------------------------------------------------------------
nh_pass_north = [
    (330, 0), (345, 8), (355, 18), (359, 28),
    (0, 35), (5, 28), (20, 18), (55, 9), (90, 0),
]
run_simulation(
    "Northern Hemisphere — satellite transits North (OTT required)",
    nh_pass_north, 'south_centered'
)

# ------------------------------------------------------------------------
# Scenario D: Northern Hemisphere — pass stays in the southern sky (no OTT)
# ------------------------------------------------------------------------
nh_pass_south = [
    (120, 0), (140, 15), (160, 30), (180, 42),
    (200, 30), (220, 15), (240, 0),
]
run_simulation(
    "Northern Hemisphere — satellite transits South (no OTT needed)",
    nh_pass_south, 'south_centered'
)
```

Sample Output

When executed, the code above produces output for all four scenarios. The two most instructive are shown below. First, the Southern Hemisphere pass that crosses south (OTT required):

Time	Sat Az	Sat El	Real Az	Real El	Method
0s	225.0°	0.0°	45.0°	180.0°	Over-the-Top
30s	210.0°	8.0°	30.0°	172.0°	Over-the-Top

60s	195.0°	18.0°	15.0°	162.0°	Over-the-Top
90s	183.0°	28.0°	3.0°	152.0°	Over-the-Top
120s	180.0°	35.0°	0.0°	145.0°	Over-the-Top
150s	177.0°	28.0°	357.0°	152.0°	Over-the-Top
180s	165.0°	18.0°	345.0°	162.0°	Over-the-Top
210s	150.0°	9.0°	330.0°	171.0°	Over-the-Top
240s	135.0°	0.0°	315.0°	180.0°	Over-the-Top

The Real Az column moves smoothly from 45° down through 0° and on to 315° — the rotator tracks through north without hesitation, while the elevation above 90° keeps the antenna correctly pointed at the satellite throughout.

Contrast this with the Southern Hemisphere pass that stays in the northern sky — no wrap-around crossing is detected, and the pass is tracked directly:

Time	Sat Az	Sat El	Real Az	Real El	Method
0s	45.0°	0.0°	45.0°	0.0°	Direct
30s	30.0°	12.0°	30.0°	12.0°	Direct
60s	10.0°	25.0°	10.0°	25.0°	Direct
90s	355.0°	40.0°	355.0°	40.0°	Direct
120s	340.0°	25.0°	340.0°	25.0°	Direct
150s	325.0°	12.0°	325.0°	12.0°	Direct
180s	315.0°	0.0°	315.0°	0.0°	Direct

Here the Sat Az and Real Az columns are identical — no transformation is applied. The rotator tracks smoothly through north (Az=355° → 315°) within its normal travel range, requiring no over-the-top intervention at all.

Practical Considerations

Several practical points are worth noting for operators who wish to implement this technique:

Elevation rotator capability: Verify that your elevation rotator physically supports movement beyond 90°. Many models designed for satellite work do, but check your controller's manual. The Yaesu G-5600B, for example, supports 0–180° of elevation travel, making it well-suited for this technique.

Software support and pre-pass planning: Implementing this technique correctly requires a pre-pass planning step, as described in the section above. If your tracking software already includes an 'allow flip' or 'over-the-top' option, verify that it performs a full pass analysis before AOS — not a reactive, on-the-fly mode switch — and that it correctly identifies your rotator type (north-centered for Southern Hemisphere stations; south-centered for Northern Hemisphere stations).

Pre-positioning before AOS: The antenna must be driven to its starting position — computed from the first predicted pass point in the decided mode — before the satellite rises. For an over-the-top pass, this means the elevation will be near 180° while the antenna waits. Ensure the controller does not interpret this as an error condition and does not reset or home the rotator while waiting for AOS.

Polarisation effects: Flipping the antenna over the top rotates the polarisation plane by 180°. For circular polarisation this is irrelevant. For linear polarisation, you may need to adjust the polarisation rotator or accept a polarisation mismatch. In practice, the signal degradation from a brief polarisation offset is generally far less severe than losing the satellite entirely due to a rotator wrap-around.

Transition hysteresis in boundary detection: When scanning the predicted azimuth track for a wrap-around crossing, include a guard margin of ±5–10° around the boundary. Any pass with AOS or LOS within this margin of the wrap boundary should be treated as requiring over-the-top mode, as the detection logic may not produce values on both sides of the boundary if the satellite rises or sets very close to it.

Conclusion

The over-the-top technique is an elegant geometric solution to a real operational problem. By exploiting the full mechanical range of an elevation rotator, satellite operators can track passes that would otherwise cause the azimuth rotator to perform a damaging, contact-breaking full rotation. The mathematics are simple, the implementation is straightforward, and the benefit — uninterrupted satellite tracking through any part of the sky — is substantial.

Whether you are in the Southern Hemisphere dealing with south-transiting passes, or in the Northern Hemisphere struggling with satellites crossing your meridian to the north, this technique applies equally. The only requirement is an elevation rotator that supports the full 0–180° range — and a tracking controller or software layer that knows when to flip.

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